On MNIST, Bayes by Backprop's OOD Advantage Runs on Optimizer Time

A softness-controlled study of mean-field VI on MNIST

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Abstract

Bayesian neural networks are often said to detect out-of-distribution (OOD) inputs better than deterministic networks. But their predictions are softer everywhere, and a deterministic network can be softened for free by a temperature. I compare each method with a deterministic network softened to the same in-distribution entropy, in a series of pre-registered experiments on MLPs trained on MNIST. At typical training budgets, mean-field Bayes by Backprop (BBB) is no better than this softness-matched network. With longer training it beats it by +0.06 to +0.08 AUROC. The advantage holds against a plain network and a 5-member deep ensemble trained for the same number of steps, and it is not reproduced by adding random weight noise to a trained network, so it depends on the variances BBB learns. Among deterministic detectors, a kNN detector on network features comes closest (within about 0.02 AUROC, and tying BBB in one setting). When this happens is set by the optimizer, not the data. Under Adam the posterior standard deviations grow on a clock of learning rate $\times$ steps: halving or doubling the learning rate doubles or halves the time, while a 20-fold change in training-set size changes it by at most 20%. Under SGD the same growth depends strongly on the training-set size, and where the standard deviations stay small BBB shows no advantage at all. Which uncertainty score is reported also matters. With the default $\mathcal{N}(0, 1)$ prior, the entropy-based advantage can disappear later in training as the model starts to underfit, while the mutual-information (MI) advantage persists. With a prior of scale 0.3 both persist. MI is also high in runs that never learned the task, so on its own it does not show a useful model. The mechanism is specific. Long training leaves BBB effectively overpruned: only 1–5% of hidden weights carry more signal than noise, and the advantage needs BBB's own means paired with its own allocation of noise. OOD images drive the noisy first-layer weights much harder relative to their signal, and a single analytic pass over the first layer's means and variances reproduces BBB's sampled OOD detection within 0.02 AUROC, confirmed on fresh models and two unseen OOD sets. A natural-gradient method (IVON) escapes the optimizer clock, its variances scaling as $1/\sqrt{N}$, and shows no advantage over the softness-matched network. On CIFAR-10 with small, underfit models (37–68% accuracy across a CNN and an MLP), no comparable advantage appears.

1 Introduction

A standard classifier gives a confident prediction for any input, including inputs unlike its training data. Bayesian neural networks (BNNs) average predictions over a distribution of weights [1], which should raise uncertainty on unfamiliar inputs. The usual evidence is that a BNN's predictive entropy is higher on OOD data than a deterministic network's.

That evidence has a confound. A BNN's averaged prediction is softer than a deterministic network's on every input, and temperature scaling [2] softens a deterministic network at no cost. Temperature is a known ingredient of OOD detection [3] and appears as a baseline in uncertainty benchmarks [4]. Whether BNNs are intrinsically good OOD detectors has been questioned [5], and a Bayesian last layer is known to reduce overconfidence far from the data [6]. The comparison I use throughout is simple: each method against a deterministic network softened, on validation data only, to that method's own in-distribution entropy. The question is whether the method ranks OOD inputs as more uncertain than in-distribution inputs better than a network of the same softness.

Findings.

1. At standard training budgets, mean-field BBB is no better than the softness-matched network on MNIST → Fashion-MNIST and KMNIST. Deep ensembles are slightly better; MC dropout ties; last-layer Laplace is worse (§3.1).

2. With enough optimizer steps BBB beats the control whatever the training-set size, and also beats step-matched plain networks and ensembles and a weight-noise control; a kNN detector on features comes within about 0.02 and ties BBB in one setting (§3.2, §3.6).

3. Under Adam, the posterior standard deviations grow on a learning-rate × steps clock that is nearly independent of the training-set size; under SGD it is not. No growth, no advantage (§3.4).

4. Along training, the entropy-based advantage can rise and fall while the MI-based advantage persists. With a better-scaled prior both persist. MI is also high for models that never learned (§3.3, §3.5).

5. The advantage needs posterior noise in both hidden layers together, and it is not a pixel shortcut (§3.6).

6. Long-trained BBB is effectively overpruned; the advantage needs its own means paired with its own per-weight noise (§3.7). OOD inputs separate in the first layer's noise-to-signal ratio, which reproduces BBB's sampled OOD detection in one analytic pass (§3.8).

7. IVON, a natural-gradient variational method, follows N rather than the step count and has no advantage over the control; inflating its variances does not create one (§3.9).

8. Neither held-out NLL nor the ELBO picks a good stopping point. A synthetic-outlier rule does no better than a fixed step count (§3.10).

9. On CIFAR-10 with small, underfit models, no comparable advantage appears (§4).

Every experiment had its hypothesis, predictions and falsification criteria written before it ran (committed to git for E1–E9; for E10–E18, SHA-256 hashes that I recorded before launch, which are self-attested rather than independently timestamped). Appendix B lists all predictions, including the many that failed.

2 Setup

Data. MNIST training subsets of $n \in \{500, 2{,}000, 10{,}000\}$ class-stratified images (one per seed); a fixed validation pool of 5,000 further training images, disjoint from every subset; the MNIST

test set; OOD: Fashion-MNIST [7] and KMNIST [8] test sets, normalised with MNIST statistics. CIFAR-10 experiments are described in §4.

Models. All MNIST models use a 784–256–256–10 ReLU MLP. *DNN*: Adam (10^{-3}, cosine), 40 epochs, batch 64. *MAP*: the same network trained to the MAP estimate under BBB's $\mathcal{N}(0,1)$ prior, with BBB's budget (150 epochs, gradient clipping). *MC dropout* [9]: $p = 0.1$, 50 samples. *Deep ensemble* [10]: 5 DNNs. *Last-layer Laplace* [11]: full generalised Gauss–Newton covariance over the last layer of a DNN, prior precision tuned on validation NLL. *IVON* [18]: the official implementation, with effective sample size N and weight decay $1/N$ (the same $\mathcal{N}(0,1)$ prior as BBB); learning rate and initial Hessian chosen once on a separate subset by validation NLL. *BBB* [1]: mean-field Gaussian posterior, $\mathcal{N}(0,1)$ prior, per-example negative ELBO

$$\mathcal{L} = \frac{1}{B} \sum_{i \in \text{batch}} -\log p(y_i \mid x_i, w) + \frac{1}{N} \mathrm{KL}\big[q(w) \,\|\, p(w)\big], \qquad w = \mu + \sigma \odot \varepsilon, \tag{1}$$

with the KL weight warmed up linearly, $\sigma = \mathrm{softplus}(\rho)$ and ρ initialised at -5 ($\sigma_0 \approx 0.007$), Adam 10^{-3}, gradient clipping, 50 posterior samples at test time. By default BBB trains for 150 epochs at $n \geq 2{,}000$ (4,800 steps at $n = 2{,}000$) and the same number of steps at $n = 500$.

The softness-matched control. For a method M, I fit a temperature T to a DNN's logits so that its mean predictive entropy on the *validation* set equals M's. No test or OOD data enters the fit. The reported quantity is $\Delta = \mathrm{AUROC}(M) - \mathrm{AUROC}(\mathrm{DNN}_T)$, where AUROC measures the separation of test from OOD images by predictive entropy. $\Delta > 0$ means M ranks OOD inputs better than a network of the same softness. The MI-based Δ uses M's mutual-information AUROC minus the same softness-matched DNN's entropy AUROC; a single deterministic network has no mutual information, so this compares two different scores and should be read as "does M's best epistemic score beat the softened network". Comparisons are paired by seed, with bootstrap confidence intervals and sign counts (wins out of seeds). Wilcoxon tests are reported only with 10 seeds, since with 5 seeds they cannot fall below $p = 0.0625$. Unless stated otherwise, BBB predictions use 50 posterior samples. The training-dynamics runs (§3.3–§3.10) use 20; I did not compare the two directly, so numbers from the two kinds of run are not interchangeable.

3 Results on MNIST

3.1 At standard budgets, only ensembles beat the control

Table 1 compares each method with its own softness-matched DNN over 10 seeds. At $n = 500$, BBB is worse than the control on both OOD sets (0/10 seeds better). At $n = 2{,}000$ it is worse on KMNIST (0/10) and borderline on Fashion-MNIST (3/10 seeds better, CI upper bound -0.001, Wilcoxon $p = 0.16$). At $n = 10{,}000$ it is much better, in all 10 seeds. §3.2 shows that this jump is not about data. Deep ensembles beat the control at every n by $+0.01$ to $+0.02$; against a stronger, budget-matched MAP network their edge mostly disappears at $n = 2{,}000$. MC dropout roughly ties the control, and last-layer Laplace is clearly below it.

On accuracy, BBB beats the budget-matched MAP network by 1.1 points at $n = 500$ and 0.5 points at $n = 2{,}000$ (9/10 and 10/10 seeds), and loses by 0.5 points at $n = 10{,}000$. About half of its apparent lead over the plain DNN at $n = 2{,}000$ is ordinary regularisation.

n	method	Δ Fashion [95% CI]	Δ KMNIST [95% CI]
500	BBB	−0.046 [−0.063, −0.032]	−0.035 [−0.041, −0.028]
500	ensemble	+0.014 [0.003, 0.027]	+0.010 [0.008, 0.012]
500	MC dropout	−0.003 [−0.016, 0.009]	−0.019 [−0.021, −0.018]
500	last-layer Laplace	−0.072 [−0.088, −0.055]	−0.112 [−0.122, −0.102]
2,000	BBB	−0.024 [−0.048, −0.001]	−0.010 [−0.014, −0.007]
2,000	ensemble	+0.015 [0.007, 0.022]	+0.015 [0.013, 0.018]
2,000	MC dropout	+0.007 [0.002, 0.012]	−0.010 [−0.012, −0.007]
2,000	last-layer Laplace	−0.040 [−0.047, −0.032]	−0.043 [−0.047, −0.038]
10,000	BBB	**+0.048** [0.038, 0.059]	**+0.069** [0.065, 0.074]
10,000	ensemble	+0.016 [0.008, 0.026]	+0.023 [0.021, 0.025]
10,000	MC dropout	−0.010 [−0.018, −0.001]	−0.007 [−0.010, −0.003]
10,000	last-layer Laplace	−0.014 [−0.023, −0.006]	−0.012 [−0.016, −0.008]

Table 1: Each method vs a DNN softened to its own validation entropy (entropy AUROC difference, 10 seeds). Source: `experiments/E2_mnist_grid`.

3.2 The advantage follows optimizer steps, not data

With fixed epochs, a larger n also means more optimizer steps: 23,550 at $n = 10{,}000$ vs 4,800 at $n \leq 2{,}000$. Figure 1 decouples the two. With the same data, 23,550 steps give $\Delta = +0.06$ to $+0.08$ in 5/5 seeds at every n, including $n = 500$. 4,800 steps give no advantage, including at $n = 10{,}000$. Over this range the training-set size barely matters for whether the advantage appears; §3.4 shows why.

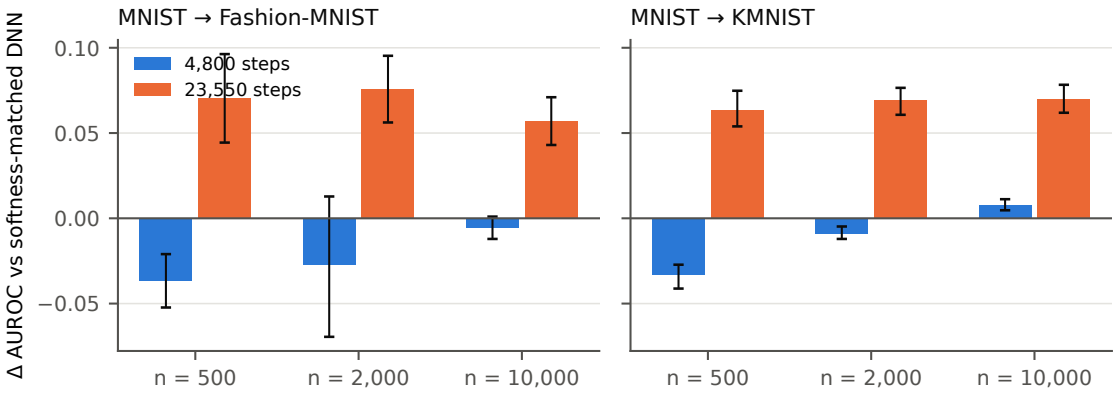


Figure 1: BBB vs its softness-matched DNN with the same training data and two step budgets (5 seeds, 95% CIs). Source: `experiments/E4_steps_vs_n`.

3.3 Along training: entropy and MI diverge

To see what happens in between, I trained BBB for 50,000 steps at $n = 2{,}000$ with a *constant* learning rate and evaluated every 1,000 steps (Figure 2). With the default initialisation, the entropy-based Δ on KMNIST is negative for the first ∼3,000 steps, peaks at +0.067 to +0.074 around 9,000–14,000 steps (mean first-layer σ 0.65–0.82), and is negative again at 25,000–30,000 steps. The MI-based Δ does not fall: it is +0.073 at 10,000 steps, +0.071 at 30,000 and +0.061 at 50,000. The

entropy advantage disappears because the model also becomes uncertain on in-distribution digits. Mean test entropy rises from 0.30 to 0.94 as accuracy falls from 0.93 to 0.89. The negative ELBO keeps improving throughout (143 at 4,000–5,000 steps, 6.8 at 24,000–25,000, 3.5 at 50,000). It drifts towards a posterior with $\sigma \approx 0.98$ in the hidden layers (0.93 in the last), accuracy 0.89 and ECE 0.19. That is consistent with Coker et al.'s analysis of wide mean-field BNNs [12], whose theorem covers a different setting (regression, odd activations). The entropy collapse is specific to $n = 2{,}000$: at 40,000 steps the entropy-based Δ is still $+0.033$ ($n = 500$) and $+0.060$ ($n = 10{,}000$) on KMNIST.

Two cautions about MI. First, a slightly larger initial σ (0.049) reaches the advantage sooner (3,000 steps instead of 6,000), but initialising $\sigma \geq 0.31$ prevents learning altogether: accuracy stays at chance in a local optimum with a far worse ELBO (9.0–9.2 vs 3.4–3.5). These chance-level models have the *largest* MI-based Δ (about $+0.10$). A positive MI Δ therefore does not by itself show a useful model. It also reflects disagreement between near-prior posterior samples on any unusual input. Second, these runs use a constant learning rate; the E4 runs (§3.2) used a cosine schedule, which freezes training wherever the budget ends.

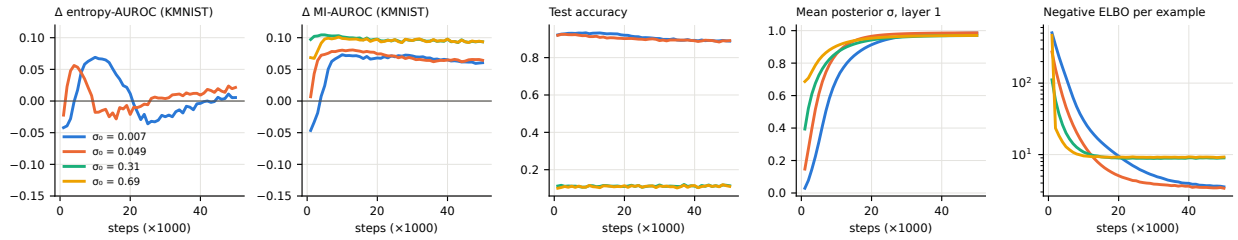


Figure 2: BBB training dynamics at $n = 2{,}000$ (constant learning rate; mean of 3 seeds per initial σ_0; 20 posterior samples). Runs with $\sigma_0 \geq 0.31$ never learn; their entropy-based Δ (about -0.86) is below the plotted range. Source: `experiments/E6_dynamics`.

3.4 An optimizer clock

Why do steps matter more than data? For one weight with posterior $\mathcal{N}(\mu, \sigma^2)$ and prior $\mathcal{N}(0, \sigma_p^2)$,

$$\mathrm{KL} = \log \frac{\sigma_p}{\sigma} + \frac{\sigma^2 + \mu^2}{2\sigma_p^2} - \frac{1}{2}, \qquad \frac{\partial\, \mathrm{KL}}{\partial \rho} = \left(\frac{\sigma}{\sigma_p^2} - \frac{1}{\sigma}\right) \mathrm{sigmoid}(\rho) \;\approx\; -1 \quad \text{for } \rho \ll 0, \tag{2}$$

because $\sigma = \mathrm{softplus}(\rho) \approx \mathrm{sigmoid}(\rho) \approx e^{\rho}$ when $\rho \ll 0$. In Eq. 1 this gradient is weighted by $1/N$, and while σ is tiny the likelihood term's gradient on ρ (which scales with σ) is negligible. So the gradient on each ρ is roughly the constant $-1/N$. Adam divides each gradient by a running estimate of its magnitude, so a gradient of constant sign produces a step of about the learning rate η, independently of its size and hence of N; this is Adam's known sign-descent behaviour [13], applied here to the variance parameters. Hence

$$t^\star \approx \frac{\rho^\star - \rho_0}{\eta}, \tag{3}$$

the number of steps for ρ to travel from ρ_0 to a target $\rho^\star$. For $\sigma = 0.3$ ($\rho^\star = -1.05$) from $\rho_0 = -5$ this gives $t^\star \approx 3{,}950$ steps at $\eta = 10^{-3}$. Under SGD the step is instead η_ρ / N (times $1/(1-\beta)$ with momentum β), so the same growth takes N times longer as N grows. The approximation breaks down once σ is no longer small compared with the prior, or once the likelihood term pulls back. Unlu and Aitchison [14] observe that mean-field variance parameters are often unconverged when

training stops, and Nishida and Taiji [15] report that BBB's variational parameters grow when it is trained with Adam. Eq. 3 makes that growth quantitative. A first hint was already in the E9 runs (Adam, learning rate 10^{-3}): the mean first-layer ρ was -3.54, -3.52 and -3.62 after 1,000 steps, -1.88, -1.85 and -1.92 after 3,000, and -1.01, -0.99 and -1.09 after 5,000, for $n = 500$, 2,000 and 10,000.

I then tested this directly (Figure 3; 3 seeds per arm, 40,000 steps, $n \in \{500, 10{,}000\}$). With Adam, the time for the mean first-layer σ to reach 0.3 was 10,000, 5,000 and 3,000 steps at $n = 500$ and 12,000, 6,000 and 3,000 at $n = 10{,}000$, for learning rates 5×10^{-4}, 10^{-3} and 2×10^{-3}. Time $\times$ learning rate was constant to within 12%, and changing n twenty-fold changed it by at most 20%. Eq. 3 predicts 7,900, 3,950 and 1,975 steps with no fitted constant: it gets the $1/\eta$ scaling and the independence from n right, and the observed times are 1.27–1.52 times longer, consistent with gradient noise making Adam's steps somewhat smaller than η. At $\eta = 10^{-3}$ the gap is larger at $n = 10{,}000$ (1.52) than at $n = 500$ (1.27), which fits that explanation: the constant $1/N$ signal is smaller relative to the minibatch noise when N is larger. These times are measured on a 1,000-step grid, which at the highest learning rate (3,000 steps) is coarser than the 12% figure; and the $n = 500$, 2×10^{-3} arm collapsed later in training (see below). The onset of the OOD advantage followed the same clock (onset $\times$ learning rate constant to within 7% at $n = 10{,}000$). With SGD (momentum 0.9) on the ρ parameters, the mean σ after 20,000 steps was 0.67 at $n = 500$ but 0.017 at $n = 10{,}000$, a 39-fold difference. In that last setting σ never grew, and BBB had *no* OOD advantage (peak Δ -0.007) while reaching the best accuracy (0.966). At $n = 500$, SGD grew σ to 0.67 by 20,000 steps but gave only a small advantage (peak Δ on KMNIST $+0.03$, never reaching the $+0.04$ onset criterion; accuracy 0.882). All four E10 predictions held; the follow-up below (E13) did less well. Two side observations: at $n = 500$ a learning rate of 2×10^{-3} collapsed training (final accuracy 0.09–0.67), and plotted against learning rate $\times$ steps the Adam curves coincide during growth, with n only shifting the plateau slightly.

In these runs gradient clipping acted on all parameters jointly, so under SGD the ρ step also depended on the cross-entropy gradient. I therefore repeated the SGD arm with clipping on the means only (3 seeds per arm). The dependence on n remained. At $n = 500$, σ reached 0.3 by 5,000 steps in every seed; at $n = 10{,}000$ it never did ($\sigma = 0.018$ at 20,000 steps, against 0.945 at $n = 500$ in the one seed that ran past 20,000 steps; the other two diverged first, see below). The SGD form of Eq. 3 predicts these numbers with no fitted constant. With $\eta_\rho = 0.05$ and momentum 0.9 the step on ρ is about $0.05 \cdot 10/N$, i.e. 5×10^{-5} at $n = 10{,}000$. After 20,000 steps (less half of the 640-step KL warm-up) that gives $\rho \approx -4.02$ and $\sigma \approx 0.018$, against 0.0177 observed (0.0173 in E10). At $n = 500$ the step is 10^{-3}, so σ should reach 0.3 after about 4,300 steps; the first checkpoint at or above 0.3 was 5,000. Without clipping, the $n = 500$ runs also reached a larger advantage ($+0.057$ to $+0.060$), so the small $+0.03$ above was partly the clipping slowing σ down. Scaling the ρ learning rate with N (1.0 at $n = 10{,}000$) made σ reach 0.3 at the same checkpoint as at $n = 500$ (5,000) in two of three seeds, but both crossed shortly before diverging (at 6,000 and 9,000 steps) and the third diverged first, so this is weak evidence. But unclipped SGD on ρ turned out to be unstable: it diverged in 6 of 9 runs, every run in which σ grew large (at 17,000–34,000 steps for $n = 500$, and 5,000–9,000 for the scaled runs). The scaled runs therefore stopped before any OOD advantage could appear (peak $+0.022$). Two of the three pre-registered predictions of this follow-up failed or could not be scored as stated.

3.5 Prior scale

The $\mathcal{N}(0, 1)$ prior is badly misspecified for layers whose weights have scale ~ 0.05 [16]. I reran the dynamics at $n = 2{,}000$ with priors of scale 0.1 and 0.3 (3 seeds, 40,000 steps). The MI-based

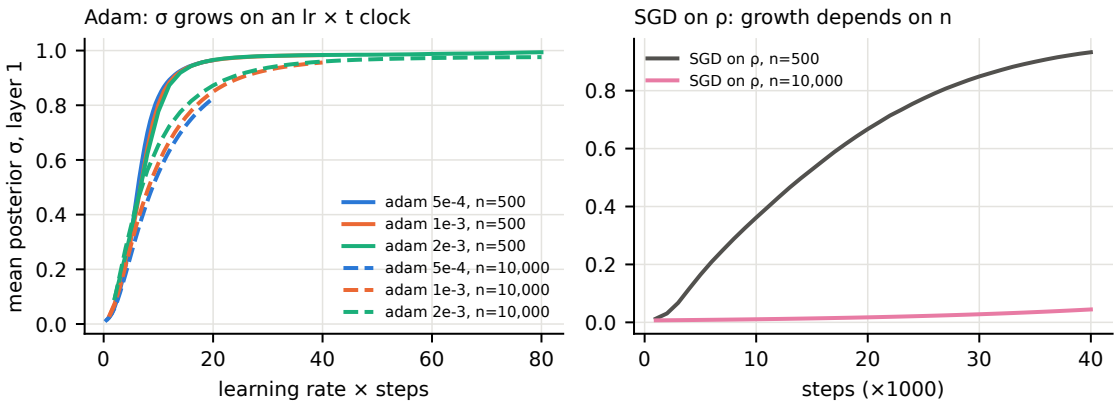


Figure 3: Mean first-layer posterior σ (mean of 3 seeds). Left: Adam at three learning rates and two training-set sizes, against learning rate × steps. Right: SGD on the ρ parameters, against steps. Source: `experiments/E10_optimizer_clock`.

advantage survived both: its peak on KMNIST was +0.055 ($\sigma_p = 0.1$) and +0.075 ($\sigma_p = 0.3$), against +0.080 for $\sigma_p = 1$. With $\sigma_p = 0.3$ the posterior σ reached the prior scale by about 10,000 steps and stayed there, and the entropy-based advantage no longer disappeared: at 40,000 steps it was +0.045 (KMNIST) and +0.048 (Fashion-MNIST), against −0.009 and +0.015 for $\sigma_p = 1$. Much of the entropy collapse in §3.3 is therefore σ drifting all the way to an oversized prior. A tighter prior did not help fit, contrary to my prediction: with $\sigma_p = 0.1$ accuracy was lower (0.870 vs 0.896), because the prior also pulls the weight means towards zero, and the Fashion-MNIST advantage mostly vanished.

3.6 What produces the advantage

Noise in both hidden layers. Taking BBB models trained for 23,550 steps, I switched posterior noise off layer by layer. Noise in the first and second layers together keeps 72–82% of Δ. Noise in any single layer keeps essentially none (first layer: 16% on Fashion-MNIST, −26% on KMNIST). The posterior-mean network on its own is a worse OOD detector than the DNN. The advantage comes from noise compounding across the hidden layers. Noise in the last layer alone does not reproduce it in this model, which may explain why last-layer Laplace did poorly in Table 1.

Not a weak baseline, and not just noise. I compared BBB trained for 23,550 steps with stronger controls (5 seeds; $n \in \{2{,}000, 10{,}000\}$; both OOD sets; Table 2). *Weight noise:* adding Gaussian noise to every layer of a trained plain network (scaled to each layer's weight spread and matched to BBB's validation entropy) gave AUROC 0.81–0.89, and BBB beat it by +0.10 to +0.15. Even the best noise scale chosen with knowledge of the OOD set stayed +0.07 to +0.10 below BBB. So the advantage is not produced by *any* weight noise; it depends on the variances BBB learns. *Matched training budgets:* a plain network and a 5-member ensemble trained for the same 23,550 steps stayed below BBB in all 5 seeds in every setting (+0.040 to +0.087 against the softened network, +0.041 to +0.075 against the ensemble). *Other scores:* on the plain networks, maximum softmax probability, maximum logit, energy and an oracle-tuned temperature reached at most 0.888–0.918 on average, and a Mahalanobis detector on penultimate features 0.86–0.94. A kNN detector on normalised features came closest, 0.93–0.97, within +0.006 to +0.023 of BBB; at $n = 10{,}000$ on Fashion-MNIST it ties BBB (2/5 seeds better).

control (vs BBB, 23,550 steps)	Δ Fashion	Δ KMNIST
plain network + weight noise (softness-matched)	+0.11 to +0.14	+0.10 to +0.15
step-matched plain network, softened	+0.040 to +0.087	+0.075 to +0.078
step-matched 5-member ensemble	+0.041 to +0.058	+0.058 to +0.075
Mahalanobis on features	+0.021 to +0.109	+0.030 to +0.059
kNN on features	+0.006 to +0.023	+0.017 to +0.023

Table 2: BBB minus each control (entropy AUROC; range over $n \in \{2{,}000, 10{,}000\}$ and, for feature detectors, two base networks; 5 seeds each). Source: `experiments/E12_controls`, `results/E12/`.

Not a pixel shortcut. Many pixels are constant (always background) across a training subset: on average 192, 148 and 104 at $n = 500$, 2,000 and 10,000. A model-free rule, "flag any image with ink on such a pixel", separates MNIST from Fashion-MNIST and KMNIST with AUROC 0.967–1.000 over 15 training subsets (means 0.990–1.000), a real weakness of these benchmarks. But the models do not rely on it. Resetting those pixels to background in the OOD images leaves every method's AUROC and Δ essentially unchanged (BBB at $n = 10{,}000$: $+0.057 \to +0.055$ and $+0.070 \to +0.070$). The growth of σ is also global, not concentrated on those pixels (ratio 1.0–1.24).

3.7 An overpruned posterior

What do BBB's long-trained variances look like? They are far from uniform (Figure 4). In the two hidden layers the median σ is 0.71–0.84, close to the prior, and only 1.2–4.5% of weights have $|\mu| > \sigma$. In the second layer $|\mu|$ and σ are strongly anti-correlated (-0.64 to -0.72), and the last layer is nearly deterministic. This is the overpruning described by Trippe and Turner [27]: a small, precise subnetwork computes the prediction and most other weights carry prior-scale noise. A pre-registered swap test shows that both halves of this pairing matter. Putting BBB's variances on the means of a plain network trained for the same number of steps collapses it to 13–15% accuracy. Keeping BBB's means but shuffling its variances within each layer cuts accuracy to 0.81 and 0.59 and turns the OOD advantage negative (-0.16 to -0.32). I had predicted that shuffling would not matter, because the *average* variance on always-blank and active pixels was similar (§3.6); that summary hid the per-weight structure.

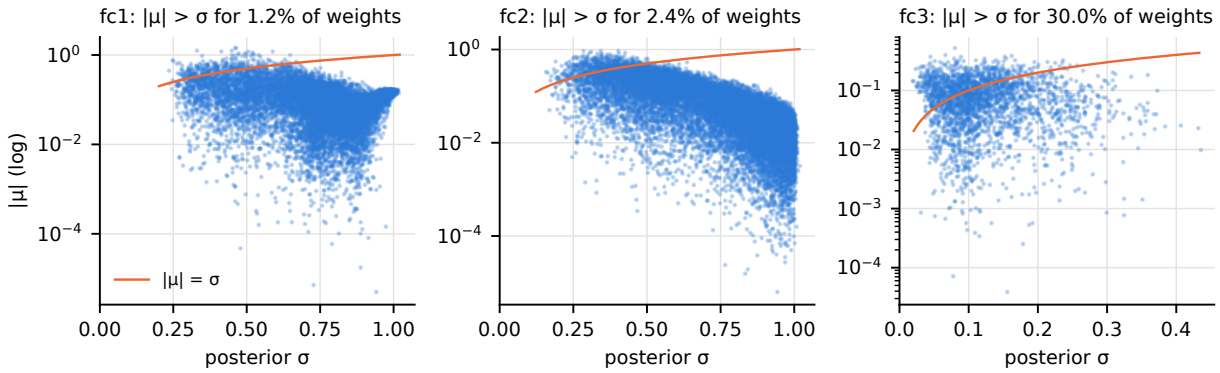


Figure 4: Posterior mean magnitude against posterior σ for every weight of a BBB network trained for 23,550 steps ($n = 2{,}000$, seed 0; 20,000 weights sampled per layer). Source: `experiments/E16_swap`.

3.8 Where OOD inputs separate: the first layer

For an input x, the first layer's mean pre-activation is $m_j = \sum_i \mu_{ij} x_i + \mu_{b_j}$ and the variance that posterior noise adds to it is $v_j = \sum_i \sigma_{ij}^2 x_i^2 + \sigma_{b_j}^2$. Their ratio,

$$R_1(x) = \frac{\sum_j v_j}{\sum_j m_j^2}, \tag{4}$$

measures how much noise an input routes through the first layer relative to the signal it produces. I had pre-registered the same ratio for the *second* layer, computed from the mean first-layer activations, as the place where in- and out-of-distribution inputs separate; that prediction failed (AUROC 0.45–0.69), partly because it ignores noise passed on from the first layer. The first-layer ratio instead matched BBB's own detection. I then confirmed this in a separate pre-registered test on ten freshly trained BBB models and on two OOD sets never used before, grayscale CIFAR-10 and SVHN (Figure 5). On all four far-OOD sets R_1 reached AUROC 0.960–0.991, within 0.003–0.019 of BBB's 50-sample mutual information, and correlated with BBB's per-image MI (Spearman 0.71–0.83). In-distribution digits are computed through the low-noise subnetwork; OOD images also drive high-variance weights. Second-layer noise is then needed to turn this into disagreement between posterior samples (§3.6). The account is incomplete for near-OOD inputs: on digits rotated by 90° at $n = 10{,}000$, BBB reached 0.950 and R_1 only 0.863, so deeper layers add information there. A practical side effect is that R_1 is a sampling-free OOD score for a trained BBB network.

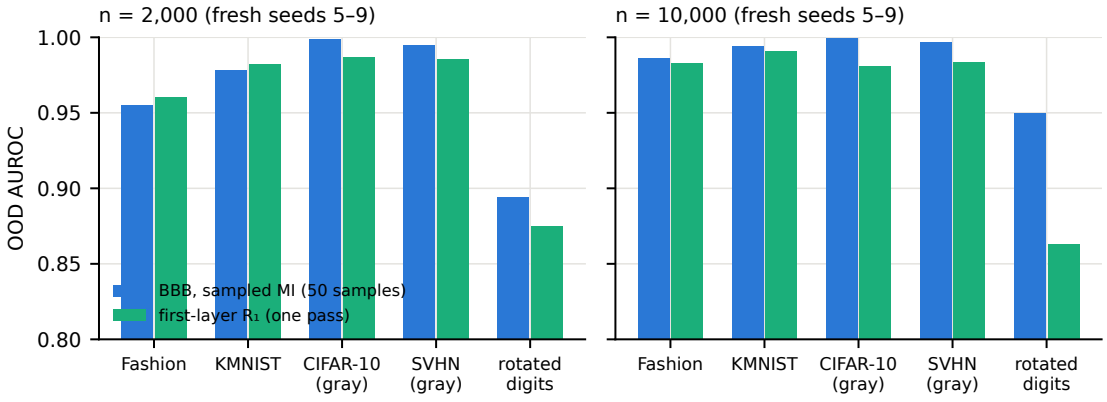


Figure 5: OOD AUROC of BBB's sampled mutual information and of the analytic first-layer ratio R_1 (Eq. 4) on freshly trained models (mean of 5 seeds). CIFAR-10 and SVHN were not used anywhere else in the MNIST study. Source: `experiments/E18_r1_confirm`.

3.9 Natural-gradient VI does not show the effect

IVON [18] keeps a Gaussian posterior $\mathcal{N}(m, 1/(N(h+\delta)))$ whose variance is set by the training-set size and a running estimate h of the loss curvature, not by an Adam step count. In a pre-registered comparison (Figure 6; 3 seeds per n, 40,000 steps), IVON's mean first-layer σ was 0.170, 0.086 and 0.039 at $n = 500$, 2,000 and 10,000: the $n = 500$ to $n = 10{,}000$ ratio of 4.41 is close to the $1/\sqrt{N}$ value of 4.47, and the variances stayed far below the prior instead of drifting to it as BBB's did (0.96–0.98). At $n \geq 2{,}000$ IVON was as accurate as the plain network (0.930 and 0.969) and its OOD behaviour did not change with training length. But it had no advantage over the softness-matched network (Δ peaks of -0.005 and -0.008; FPR95 no better), against BBB's $+0.07$ and

+0.065. At $n = 500$ IVON was less accurate than the plain network and its Δ drifted down. To test whether variance size alone explains BBB's advantage, I then inflated IVON's variances by tempering the likelihood ($p(D \mid w)^{1/k} p(w)$, $k \in \{10, 100\}$, same prior). This raised σ by 3.1–9.3×, but no tempered model gained an advantage: the only one that still learned ($n = 10{,}000$, $k = 10$, accuracy 0.951) stayed at −0.004, and the others collapsed to chance accuracy. Together with the weight-noise control and the swap test, this points to the specific pairing BBB learns, not to the size of its variances. One caveat: IVON's learning rate and initial Hessian were tuned for $k = 1$ and kept fixed, so the collapses may partly reflect settings unsuited to tempering.

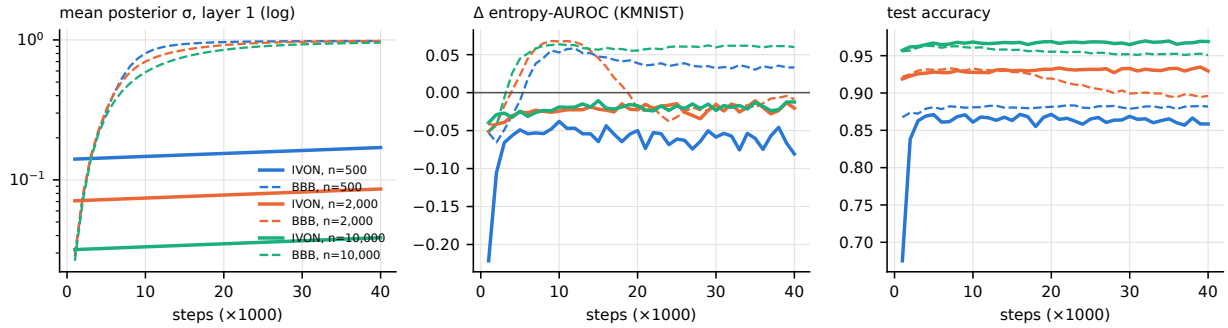


Figure 6: IVON (solid) and BBB under Adam (dashed) at three training-set sizes (mean of 3 seeds; constant learning rate). Left: mean first-layer posterior σ (log scale). Middle: entropy-AUROC difference from the softness-matched network on KMNIST. Right: test accuracy. Source: `experiments/E14_ivon`.

3.10 Choosing when to stop

Neither in-distribution criterion picks a good checkpoint. Minimum held-out NLL stops at 1,000–3,000 steps, before the advantage appears; minimum −ELBO stops at the end of training. I tested a rule that needs no real OOD data. I built three synthetic outlier sets from the validation images (uniform noise, flipped digits, pixel-permuted digits) and picked the checkpoint that best separates validation images from them by MI (Table 3; 12 runs, $n \in \{500, 2{,}000, 10{,}000\}$). It reaches 74% of the best achievable entropy-based advantage (the "oracle"), but simply stopping at a fixed 8,000 steps reaches 72%. Given the clock in §3.4, a fixed step count is a sensible rule for a fixed learning rate. Scoring the last checkpoint by MI instead of entropy, with no selection at all, does as well (+0.049). The oracle is a maximum over 40 checkpoints and is optimistic. These runs also share configurations and seeds with §3.3. The proxy rule therefore does not beat trivial alternatives here.

rule	mean step	$\bar{\Delta}$ (entropy)	fraction of oracle
oracle (uses real OOD; optimistic)	10,917	+0.067	1.00
synthetic-outlier proxies (MI)	10,417	+0.049	0.74
fixed stop at 8,000 steps	8,000	+0.048	0.72
fixed stop at 10,000 steps	10,000	+0.047	0.71
minimum −ELBO	39,417	+0.034	0.51
minimum validation NLL	2,250	−0.032	−0.48

Table 3: Stopping rules ($\bar{\Delta}$ = mean of Fashion-MNIST and KMNIST entropy-based Δ at the chosen checkpoint; 12 runs; fraction = mean $\bar{\Delta}$ of the rule / mean $\bar{\Delta}$ of the oracle). Source: `experiments/E9_selection`, `results/E9/`.

4 CIFAR-10

I repeated the dynamics experiment on CIFAR-10 ($n = 10{,}000$, no augmentation) with a small CNN (three 3×3 conv layers of 32, 64 and 128 channels, then 256 and 10 units), every layer Bayesian. OOD sets were SVHN (10,000 images) and CIFAR-100. No comparable advantage appeared (Figure 7). BBB's peak Δ on SVHN was −0.044, +0.006 and −0.056 in the three seeds, and on CIFAR-100 at most +0.031, early in training. Even a 5-member deep ensemble lost to a single softness-matched DNN on SVHN (AUROC 0.727–0.740 vs 0.759–0.807), while winning on CIFAR-100 by +0.02 to +0.03. Scored by MI instead, BBB was *worse* on SVHN: MI-AUROC ranged from 0.24 to 0.60 across checkpoints, often below chance. The drift towards the prior did replicate. The large fully connected layer's σ reached 0.996, the ELBO kept improving, and accuracy fell from 0.67–0.68 to 0.61–0.63. The first conv layer's σ stayed small (0.04).

To separate architecture from data, I trained the MNIST study's MLP on flattened CIFAR-10 images at $n = 2{,}000$ and 10,000 (3 seeds each). Its deterministic version reaches 38% and 46% test accuracy. BBB again showed no advantage on SVHN (peak Δ −0.031 to +0.030). On CIFAR-100 it gained +0.030 to +0.046 for the first few thousand steps, similar to the CNN's +0.024 to +0.031. Then it collapsed towards the prior (first-layer $\sigma \to 0.997$–0.999, accuracy falling to 0.10–0.29). Much larger CIFAR-100 "gains" at $n = 2{,}000$ (+0.09 to +0.21) appear only after collapse, at near-chance accuracy, and are not meaningful. My pre-registered decision rule counted these runs as support for an architectural explanation, but it had no accuracy condition, and the qualifying gains are barely above its threshold and no larger than the CNN's. On natural images, neither architecture shows the MNIST-sized advantage (+0.06 to +0.08 on both OOD sets). The evidence points more to the data, or to the much harder task (37–47% accuracy), than to the CNN's structure.

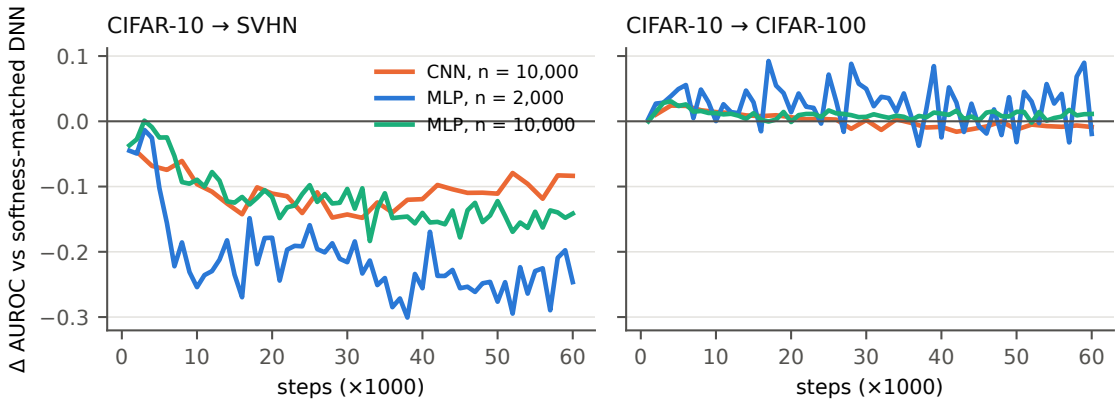


Figure 7: BBB vs its softness-matched network on CIFAR-10 over training (mean of 3 seeds per curve; CNN from all layers Bayesian, MLP = the MNIST architecture on flattened images). Sources: `experiments/E7_cifar`, `experiments/E7b_cifar_mlp`.

5 Related work

Coker et al. [12] prove that the optimal mean-field posterior predictive approaches the prior predictive as width grows; the drift in §3.3 is consistent with that effect along the optimisation path of a standard-width ReLU classifier, a setting their theorem does not cover. Unlu and Aitchison [14] note that mean-field variance parameters are often unconverged when training stops and recommend a larger learning rate for them; Nishida and Taiji [15] report that BBB's variational parameters grow under Adam and propose a decoupled optimizer (AdamB). §3.4 derives the N-independent clock behind this growth and links it to when BBB's OOD advantage appears. In natural-gradient variational methods such as Vadam [17] and IVON [18] the variances are tied to N and the loss curvature by construction; §3.9 confirms that IVON does not follow the clock, and also shows that it does not reproduce BBB's OOD advantage. Foong et al. [19] show what mean-field and MC dropout approximations cannot represent. Wenzel et al. [20] show that tempered ("cold") posteriors can predict better than the Bayes posterior; the softness-matched control here asks a related question about temperature, for OOD detection. Depeweg et al. [21] decompose predictive uncertainty into aleatoric and epistemic (MI) parts, the split behind §3.3; Wimmer et al. [22] question whether conditional entropy and MI are appropriate measures of these parts, which fits the caution about MI in §3.3. Shen et al. [18] show that a natural-gradient variational method (IVON) trains large networks as well as Adam; whether its variances follow a similar clock is open. Izmailov et al. [23] show that weights on always-zero pixels keep their prior under HMC and hurt accuracy under covariate shift; I find no such pixel-specific effect in BBB's OOD detection. Ovadia et al. [4] include temperature-scaled baselines and find ensembles strongest under shift; Ashukha et al. [24] argue that uncertainty methods should be compared after temperature scaling, which the softness-matched control extends to OOD detection; and Abe et al. [25] find that ensembles' OOD gains are largely explained by their in-distribution performance. Odgers et al. [26] show that mean-field VI can overestimate predictive variance on in-distribution points, which matches the rising in-distribution entropy in §3.3. Trippe and Turner [27] describe overpruning, where variational BNNs switch off units by sending their weights to the prior; §3.7 finds the same structure in long-trained BBB and links it to OOD detection. Fortuin et al. [16] show that isotropic Gaussian priors are often poor choices for neural networks. Henning et al. [5] question whether exact BNN posteriors are intrinsically good OOD detectors. Raina and Schmah [28] report Bayesian advantages for OOD detection from small

training sets; the results here suggest checking training length and softness before attributing such advantages to the posterior.

6 Limitations

- **Scope.** The positive results come from small MLPs on MNIST-family data. The CIFAR-10 models were small and underfit (37–68% accuracy, no augmentation), so the CIFAR result says little about well-trained networks.
- **Prior.** Only three prior scales (0.1, 0.3, 1) at one training-set size; per-layer or fan-in-scaled priors were not tried.
- **Schedule and samples.** The dynamics runs use a constant learning rate and 20 posterior samples; the fixed-budget runs use a cosine schedule and 50 samples.
- **Statistics.** Dynamics results rest on 3 seeds per setting, and the selection analysis reuses those configurations and seeds.
- **The clock.** Tested with Adam and SGD on one architecture; other adaptive optimizers and natural-gradient methods such as IVON [18] were not tried. Unclipped SGD on ρ diverged whenever σ grew large, so the SGD contrast rests on the stable phase.
- **Baselines.** All methods use the same small MLP with default hyperparameters, without per-method tuning. A kNN feature detector comes within 0.02 of BBB and ties it in one setting.
- **Mechanism.** The first-layer account was found in an exploratory analysis and then confirmed in a separate pre-registered test; it does not cover near-OOD inputs. IVON was run with one tuned setting, kept fixed when tempering.
- **No exact posterior.** Without HMC or similar, I cannot say whether any of this reflects Bayesian behaviour or only mean-field VI under Adam.

7 Conclusion

For Bayes by Backprop on MNIST-family data, "knowing what it doesn't know" is not a fixed property of the method. It is set by optimizer time. Under Adam the posterior variances grow on a learning-rate $\times$ steps clock, almost regardless of how much data there is, and the OOD advantage switches on with them. Where the variances stay small, as under SGD with more data, there is no advantage. At typical budgets BBB therefore behaves like a temperature-scaled network; with longer training it beats the softness-matched control, step-matched plain networks and ensembles, and a weight-noise control, with a kNN feature detector close behind. The uncertainty score and the prior scale then decide whether the advantage lasts. Mechanistically, long training leaves BBB overpruned, and OOD inputs route disproportionate noise through its first layer; a natural-gradient method that ties its variances to the data instead shows no such advantage. Comparisons between BBB and other methods should report the optimizer, learning rate and training budget, the score, and a softness-matched baseline.

Reproducibility. Code: `bnnstudy/`; every command: `COMMANDS.md`; every number in this report: `CLAIMS.md`; pre-registered protocols with results: `experiments/`.

A KL scaling

Dividing the KL term in Eq. 1 by the number of mini-batches instead of by N is an easy slip when the cross-entropy is averaged per batch. It over-weights the KL term by $N/\lceil N/B \rceil \approx B$ (62.5× at $N = 2{,}000$, $B = 64$). In a 3-seed pilot it cost BBB 3.2 points of accuracy (0.904 vs 0.936) and raised ECE from 0.022 to 0.185.

B Pre-registered predictions and outcomes

Each protocol (`experiments/<id>/protocol.md`) was written before its runs: committed to git for E1–E9, and for E10–E18 with SHA-256 hashes I recorded before launch (`experiments/E10_E11_prereg.txt`, `experiments/E12_E13_prereg.txt`, `experiments/E14_prereg.txt` to `E18_prereg.txt`; self-attested, not independently timestamped). Of the 61 entries below, 26 were confirmed, 24 falsified, 10 had mixed outcomes and 1 had no outcome. (E5's P1 held by construction and is omitted.) The falsified ones are what moved the study from "is the advantage just softness?" to the training-path account.

exp.	prediction	outcome
E1	P0: new pipeline reproduces old numbers (±0.02 acc, ±0.03 AUROC)	acc/ECE yes; AUROC off by 0.035
E1	P1: correct KL weight raises BBB accuracy	confirmed
E1	P2: BBB within 0.02 AUROC of softness-matched DNN	falsified (Fashion)
E2	P1: BBB never beats the control	falsified ($n = 10{,}000$)
E2	P2: ensembles beat the control	confirmed
E2	P3: MC dropout and Laplace within 0.02 of the control	dropout yes; Laplace falsified
E2	P4: larger KL weight lowers accuracy, raises KMNIST MI-AUROC	confirmed
E3	P1: BBB and budget-matched MAP equally accurate (< 0.005)	falsified (narrowly)
E3	P2: BBB never beats softness-matched MAP	falsified ($n = 10{,}000$)
E4	P1: the gain follows optimizer steps, not n	confirmed
E4	P2: dead-pixel weights have $> 5\times$ the σ of live ones	falsified
E4	P3: $n = 500$ with long training shows the gain	confirmed
E5	P2: removing dead-pixel ink halves BBB's gain	falsified
E5	P3: removing it shrinks the ensemble's gain	falsified
E5	P4: the DNN's AUROC drops less than BBB's when the ink is removed	falsified (neither drops)
E6	P1: Δ rises with σ	falsified (peaks, then falls)
E6	P2: Δ plateaus	falsified (transient)
E6	P3: larger σ_0 reaches the gain sooner	falsified ($\sigma_0 \geq 0.31$ never learns)
E6	P4: ELBO still improving at 4,800 steps	confirmed
E6b	P1: first-layer noise alone keeps $\geq 70\%$	falsified
E6b	P2: last-layer noise alone keeps $< 30\%$	confirmed
E9	P1: combined proxies reach $\geq 70\%$ of oracle; NLL $\leq 20\%$	confirmed, but a fixed 8k-step stop does as well
E9	P2: one proxy alone reaches $\geq 70\%$	falsified (67%)
E9	P3: peak step rises with n	confirmed by mean; not robust
E7	P1: window on SVHN with a CNN	falsified
E7	P2: window closes	no outcome (no window)
E7	P3: proxy selection beats NLL	falsified

exp.	prediction	outcome
E7	P4: ensemble beats the control on SVHN	falsified
E7b	rule: architecture vs data	"architecture" by the rule; weak
E10	P1: time to $\sigma = 0.3$ × learning rate constant ($\pm 30\%$)	confirmed
E10	P2: under Adam, n changes that time by $\leq 25\%$	confirmed
E10	P3: under SGD, σ grows $\geq 3\times$ faster at $n = 500$ than 10,000	confirmed ($39\times$)
E10	P4: onset of the OOD advantage follows the same clock ($\pm 40\%$)	confirmed
E11	P1: a tighter prior improves final accuracy	falsified
E11	P2: a tighter prior keeps the entropy advantage to the end	$\sigma_p = 0.3$ yes; 0.1 no
E11	P3: MI advantage survives $\sigma_p \in \{0.1, 0.3\}$ (kill test)	confirmed
E12	P1: weight noise stays ≥ 0.02 below BBB	confirmed
E12	P2: step-matched softened network stays ≥ 0.04 below BBB	confirmed (narrowly)
E12	P3: step-matched ensemble stays below BBB	confirmed
E12	P4: feature detectors ≥ 0.95 and within 0.03 of BBB	kNN yes; Mahalanobis no
E13	P1: without joint clipping, SGD σ still grows $\geq 3\times$ faster at $n = 500$	confirmed ($53\times$, one $n = 500$ seed past 20k)
E13	P2: ρ learning rate $\propto N$ aligns the clocks across n	falsified as scored (1 run diverged first; 2 crossed just before diverging)
E13	P3: with that scaling, the OOD advantage appears at $n = 10{,}000$	falsified (runs diverged first)
E14	P1: IVON's σ is $\geq 2\times$ larger at $n = 500$ than 10,000	confirmed ($4.41\times$)
E14	P2: IVON's $\sigma \leq 0.3$ at $n \geq 2{,}000$	confirmed
E14	P3: IVON shows no transient	$n \geq 2{,}000$ yes; $n = 500$ no
E14	P4: IVON's advantage below BBB's but ≥ 0	below BBB yes; ≥ 0 no
E14	P5: IVON accuracy within 0.01 of the plain network	$n \geq 2{,}000$ yes; $n = 500$ no
E15	P1: tempering raises IVON's σ	confirmed
E15	P2: at $k = 100$, IVON gains an advantage ≥ 0.03	falsified (collapse)
E15	P3: at $k = 10$, accuracy within 0.02	$n = 10{,}000$ yes; $n = 2{,}000$ no
E16	P1: BBB's σ on plain-network means collapses accuracy	confirmed
E16	P2: that swap loses ≥ 0.05 of the advantage	confirmed
E16	P3: shuffling σ within layers keeps $\geq 80\%$	falsified
E17	P1: second-layer ratio R_2 separates OOD (AUROC $\geq$ 0.90)	falsified
E17	P2: R_2 correlates with MI (≥ 0.6)	falsified
E17	P3: R_2 separates better than R_1	falsified
E18	P1: R_1 AUROC ≥ 0.90 on four far-OOD sets (fresh models)	confirmed
E18	P2: R_1 within 0.03 of BBB's MI	confirmed
E18	P3: Spearman(R_1, MI) ≥ 0.6	confirmed
E18	P4: R_1 within 0.05 of MI on rotated digits	$n = 2{,}000$ yes; $n = 10{,}000$ no

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